

SAFETY DATA SHEET

Classified in accordance with 29 CFR 1910.1200

1. Identification

Product identifier: Hydrocarbons C5 Rich**Other means of identification**

Common name(s), Crude Isoprene; Isoprene/Piperylene Concentrate; C5 Mixture; Isoprene
synonym(s): Product
SDS number: NOVA-0007

Recommended use and restriction on use**Recommended use:** Petrochemical feedstock.**Restrictions on use:** All uses other than the identified.**Manufacturer/Importer/Supplier/Distributor Information****Importer**

Company Name: NOVA Chemicals, Inc.
Address: 1555 Coraopolis Heights Road
Moon Township, PA, USA 15108
Telephone: Product Information: 1-412-490-4063
SDS Information Email: msdsemail@novachem.com

Emergency telephone number:

1-800-561-6682, 1-403-314-8767 (NOVA Chemicals) (24 hours)
1-800-424-9300 (CHEMTREC) (24 hours)

2. Hazard(s) identification

Hazard Classification**Physical Hazards**

Flammable liquids Category 1

Health Hazards

Acute toxicity (Oral) Category 4
Skin Corrosion/Irritation Category 2
Serious Eye Damage/Eye Irritation Category 2
Germ Cell Mutagenicity Category 2
Carcinogenicity Category 1B
Specific Target Organ Toxicity -
Single Exposure Category 3
Aspiration Hazard Category 1

Environmental Hazards

Acute hazards to the aquatic
environment Category 3
Chronic hazards to the aquatic
environment Category 3

Label Elements

Hazard Symbol:**Signal Word:**

Danger

Hazard Statement:

Extremely flammable liquid and vapor.
Harmful if swallowed.
Causes skin irritation.
Causes serious eye irritation.
Suspected of causing genetic defects.
May cause cancer.
May cause drowsiness or dizziness.
May be fatal if swallowed and enters airways.
Harmful to aquatic life with long lasting effects.

Precautionary Statements:**Prevention:**

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Keep container tightly closed. Ground and bond container and receiving equipment. Use explosion-proof electrical, ventilating and lighting equipment. Use non-sparking tools. Take action to prevent static discharges. Avoid breathing vapors. Wash face, hands and any exposed skin thoroughly after handling. Do not eat, drink or smoke when using this product. Use only outdoors or in a well-ventilated area. Wear protective gloves/ protective clothing/ eye protection/ face protection. Avoid release to the environment.

Response:

IF SWALLOWED: Immediately call a POISON CENTER. Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing and wash it before reuse. Rinse skin with water [or shower]. If skin irritation occurs: Get medical advice. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER if you feel unwell. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention. IF exposed or concerned: Get medical attention. In case of fire: Use dry chemical, foam, carbon dioxide (CO₂), water spray or fog to extinguish.

Storage:

Store in a well-ventilated place. Keep container tightly closed. Keep cool. Store locked up.

Disposal:

Dispose of contents and container in accordance with local regulations.

Other hazards which do not result in GHS classification:

Static accumulating flammable liquid can become electrostatically charged even in bonded and grounded equipment. Sparks may ignite liquid and vapor. May cause flash fire or explosion.

3. Composition/information on ingredients

Mixtures

Chemical Identity	Common name and synonyms	CAS number	Content in percent (%) [*]
1,3-Pentadiene	Piperylene	504-60-9	19 - 45%
Cyclopentene	1-Cyclopentene	142-29-0	9 - 25%
1-Pentene	Pent-1-ene	109-67-1	5 - 25%
1,3-Butadiene, 2-methyl-	Isoprene	78-79-5	5 - 20%
2-Pentene	Pent-2-ene	109-68-2	5 - 10%
1,3-Cyclopentadiene	Cyclopentadiene	542-92-7	0 - 3%
n-Pentane	Pentane	109-66-0	0 - 3%
Cyclopentane	Pentamethylene	287-92-3	0.1 - 2.5%
2-Butene, 2-methyl-	Amylene	513-35-9	0.5 - 2%
Butane, 2-methyl-	2-Methylbutane, Isopentane	78-78-4	0 - 2%
4,7-Methano-1H-indene, 3a,4,7,7a-tetrahydro-	Dicyclopentadiene, DCPD	77-73-6	0 - 2%
Pentane, 2-methyl-	Isohexane	107-83-5	0.1 - 1.5%

^{*} All concentrations are percent by weight.

Additional Information:

This product has been assigned a CAS# of 68476-55-1 - Hydrocarbons, C5 Rich. Benzene (CAS # 71-43-2) is also present up to 0.09 wt%. This product is inhibited with BHT (butylated hydroxytoluene) (50 to 200 ppm) for storage and transportation. This product is considered hazardous by the OSHA Hazard Communication Standard, (29 CFR 1910.1200).

4. First-aid measures

Inhalation:

IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER if you feel unwell.

Ingestion:

IF SWALLOWED: Immediately call a POISON CENTER. Rinse mouth. Do NOT induce vomiting.

Skin Contact:

IF ON SKIN (or hair): Take off immediately all contaminated clothing and wash it before reuse. Rinse skin with water [or shower]. If skin irritation occurs: Get medical advice.

Eye contact:

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention.

Most important symptoms/effects, acute and delayed

Symptoms:

Skin irritation. Eye irritation. Vomiting, abdominal pain and central nervous system effects including headache, dizziness and nausea.

Indication of immediate medical attention and special treatment needed

Treatment:

Ensure thorough eye and skin decontamination. Treat unconsciousness, nausea, hypotension, seizures and cardiac rhythm disturbance in the conventional manner. Aspiration of this product during induced emesis can result in lung injury. If evacuation of stomach contents is considered necessary, use the method least likely to cause aspiration, such as gastric lavage after protecting the airway. Observe hospitalized patients for delayed chemical pneumonia, acute kidney failure, encephalopathy, and cardiac rhythm

disturbance. Adrenergic (epinephrine, norepinephrine) and dopaminergic agonists should be avoided during treatment or used with caution (lowest effective dose) because of possible cardiac sensitization by this product mixture.

5. Fire-fighting measures

General Fire Hazards: Extremely flammable liquid and vapor. Vapors are heavier than air and may travel to a source of ignition and flash back. Closed containers may rupture violently when heated. Material will float and can be re-ignited on surface of water. If tank, rail car or tank truck is involved in fire, ISOLATE for 800 meters (1/2 mile) in all directions; also, consider initial evacuation for 800 meters (1/2 mile) in all directions. Vapors may form explosive mixture with air. Keep containers away from source of heat or fire. This product may be a static accumulator which can form an ignitable vapor-air mixture in a storage tank.

Suitable (and unsuitable) extinguishing media

Suitable extinguishing media: Use dry chemical, foam, carbon dioxide (CO₂), water spray or fog to extinguish. Use water to cool fire-exposed containers and to protect personnel.

Unsuitable extinguishing media: Do not use straight/direct streams as this may actually spread flames.

Specific hazards arising from the chemical: Upon combustion, this product emits carbon monoxide, carbon dioxide, low molecular weight hydrocarbons.

Special protective equipment and precautions for fire-fighters

Special fire-fighting procedures: Keep upwind. Keep unauthorized personnel away. Move containers from fire area if you can do so without risk. Fight fire from maximum distance or use unmanned holders or monitor nozzles. Immediately withdraw in case of fire and container venting or heat discoloration of a container. Avoid inhaling any smoke and combustion materials. Remove and isolate contaminated clothing and shoes. Cool containers with flooding quantities of water until well after the fire is out. Prevent runoff from fire control or dilution from entering streams, sewers, or drinking water supply. Reference Emergency Response Guidebook No. 128 for additional details and instructions.

Special protective equipment for fire-fighters: Wear positive pressure self-contained breathing apparatus (SCBA).

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures: Wear appropriate personal protective equipment. Isolate area. Keep unauthorized personnel away. Alert stand-by emergency and fire-fighting personnel. Monitor surrounding area for buildup of flammable concentrations in air.

Methods and material for containment and cleaning up: Do not touch or walk through spilled material. In case of leakage, eliminate all ignition sources. As an immediate precautionary measure, isolate spill or leak area for at least 50 meters (164 feet) in all directions. Keep upwind. Keep out of low areas. Stop leak if safe to do so. Contain discharge by booming on water or diking on ground. Prevent runoff from fire control or dilution from entering streams, sewers, or drinking water supply.

Small Spills: Remove liquid material with non-sparking approved pumps, skimmers or vacuum equipment. Absorb with earth, sand or other non-

combustible material and transfer to containers for later disposal. Product may react (exothermic, ignite, polymerize) with some absorbents. Test with small quantity prior to using larger amounts. Use non-sparking tools.

Large Spills: Consider downwind evacuation for 300 meters (1000 feet). Dike far ahead of larger spills for later disposal. Spills on water will volatilize rapidly, making containment or recovery difficult. A vapor-suppressing foam may be used to reduce vapors. Remove pooled liquid material with approved, non-sparking pumps, skimmers or vacuum equipment. Absorb or cover with dry earth, sand or other non-combustible material and transfer to containers. Product may react (exothermic, ignite, polymerize) with some absorbents. Test with small quantity prior to using larger amounts. Soil remediation may be required.

7. Handling and storage

Precautions for safe handling:

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Keep container tightly closed. Ground and bond container and receiving equipment. Use explosion-proof electrical, ventilating and lighting equipment. Use non-sparking tools. Take action to prevent static discharges. These alone may be insufficient to remove static electricity. For additional information on equipment bonding and grounding, refer to the American Petroleum Institute (API) Recommended Practice 2003, "Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents" or National Fire Protection Association (NFPA) 77, "Recommended Practice on Static Electricity". For additional information on storing and handling flammable liquids, refer to the National Fire Protection Association (NFPA) 30, "Flammable and Combustible Liquids Code". Maintain adequate inhibitor concentrations, and nitrogen pad bulk shipping containers to keep oxygen in headspace below 100 ppm. Deposits of organic buildup in lines or process equipment may react spontaneously with air, causing smoking and possibly a fire. Take special precautions when cold cutting or breaking into lines, or when cleaning and disposing of empty containers. Avoid breathing vapor. Avoid contact with skin and eyes. Keep away from incompatible materials such as oxidizing agents and acids. Wash face, hands and any exposed skin thoroughly after handling. Do not eat, drink or smoke when using this product. Use only outdoors or in a well-ventilated area. Wear protective gloves/ protective clothing/ eye protection/ face protection. In case of inadequate ventilation, use respiratory protection. Avoid release to the environment.

Conditions for safe storage, including any incompatibilities:

Storage area should be clearly identified, well-illuminated and clear of obstruction. Maintain adequate inhibitor concentrations, and nitrogen pad bulk shipping containers to keep oxygen in headspace below 100 ppm. Store in a well-ventilated place. Keep container tightly closed. Keep cool. Store locked up. Only allow access to authorized persons. Store and handle in properly designed pressure vessels and equipment. Store and use away from heat, sparks, open flame, or any other ignition source. Use non-sparking ventilation systems, approved explosion-proof equipment, and intrinsically safe electrical systems. Have appropriate extinguishing capability in storage area (e.g. sprinkler system, portable fire extinguishers) and flammable gas detectors. Keep absorbents for leaks and spills readily available. Inspect vents during winter conditions for vapor ice buildup. Storage tanks should be above ground and diked to hold entire contents. Store away from incompatible materials. Store according to applicable regulations and standards for flammable materials.

This product is inhibited with BHT (butylated hydroxytoluene) (50 to 200 ppm) for storage and transportation.

8. Exposure controls/personal protection**Control Parameters****Occupational Exposure Limits**

1,3-Butadiene, 2-methyl-: The American Industrial Hygiene Association (AIHA) has developed an 8-hour TWA Workplace Environmental Exposure Level (WEEL) for isoprene of 2 ppm; adoption of this WEEL is recommended.

Components	Type	Exposure Limit Values		Source
1,3-Cyclopentadiene	REL	75 ppm	200 mg/m3	US. NIOSH: Pocket Guide to Chemical Hazards, as amended
	IDLH	750 ppm		US. NIOSH. Immediately Dangerous to Life or Health (IDLH) Values, as amended
	PEL	75 ppm	200 mg/m3	US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000), as amended
	TWA	75 ppm	200 mg/m3	US. OSHA Table Z-1-A (29 CFR 1910.1000), as amended
	STEL	1 ppm		US. ACGIH Threshold Limit Values, as amended
n-Pentane	TWA	0.5 ppm		US. ACGIH Threshold Limit Values, as amended
	REL	120 ppm	350 mg/m3	US. NIOSH: Pocket Guide to Chemical Hazards, as amended
	Ceil_Time	610 ppm	1,800 mg/m3	US. NIOSH: Pocket Guide to Chemical Hazards, as amended
	IDLH	1,500 ppm		US. NIOSH. Immediately Dangerous to Life or Health (IDLH) Values, as amended
	PEL	1,000 ppm	2,950 mg/m3	US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000), as amended
	STEL	750 ppm	2,250 mg/m3	US. OSHA Table Z-1-A (29 CFR 1910.1000), as amended
	TWA	600 ppm	1,800 mg/m3	US. OSHA Table Z-1-A (29 CFR 1910.1000), as amended
	TWA	1,000 ppm		US. ACGIH Threshold Limit Values, as amended
Cyclopentane	REL	600 ppm	1,720 mg/m3	US. NIOSH: Pocket Guide to Chemical Hazards, as amended
	TWA	600 ppm	1,720 mg/m3	US. OSHA Table Z-1-A (29 CFR 1910.1000), as amended
	TWA	1,000 ppm		US. ACGIH Threshold Limit Values, as amended
2-Butene, 2-methyl-	TWA	10 ppm		US. ACGIH Threshold Limit Values, as amended
Butane, 2-methyl-	TWA	1,000 ppm		US. ACGIH Threshold Limit Values, as amended
4,7-Methano-1H-indene, 3a,4,7,7a-tetrahydro-	REL	5 ppm	30 mg/m3	US. NIOSH: Pocket Guide to Chemical Hazards, as amended
	TWA	5 ppm	30 mg/m3	US. OSHA Table Z-1-A (29 CFR 1910.1000), as amended
	TWA	0.5 ppm		US. ACGIH Threshold Limit Values, as amended
	STEL	1 ppm		US. ACGIH Threshold Limit Values, as amended
Pentane, 2-methyl-	Ceil_Time	510 ppm	1,800 mg/m3	US. NIOSH: Pocket Guide to Chemical Hazards, as amended
	REL	100 ppm	350 mg/m3	US. NIOSH: Pocket Guide to Chemical Hazards, as amended
	STEL	1,000 ppm	3,600 mg/m3	US. OSHA Table Z-1-A (29 CFR 1910.1000), as amended
	TWA	500 ppm	1,800 mg/m3	US. OSHA Table Z-1-A (29 CFR 1910.1000), as amended

	TWA	200 ppm		US. ACGIH Threshold Limit Values, as amended
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Please refer to the latest edition of the appropriate source text and consult an industrial hygienist or similar professional, or local agencies, for further information.

Appropriate Engineering Controls

Engineering methods to reduce hazardous exposure are preferred controls. Methods include mechanical ventilation (dilution and local exhaust) process or personal enclosure, remote and automated operation, control of process conditions, leak detection and repair systems, and other process modifications. Ensure all exhaust ventilation systems are discharged to outdoors, away from air intakes and ignition sources. Supply sufficient replacement air to make up for air removed by exhaust systems. Administrative (procedure) controls and use of personal protective equipment may also be required.

Individual protection measures, such as personal protective equipment (PPE)

General information:	Personal protective equipment (PPE) should not be considered a long-term solution to exposure control. Employer programs to properly select, fit, maintain and train employees to use equipment must accompany PPE. Consult a competent industrial hygiene resource, the PPE manufacturer's recommendation, and/or applicable regulations to determine hazard potential and ensure adequate protection.
Eye/face protection:	Safety glasses. Chemical goggles are recommended if splashing is possible or to prevent eye irritation from vapors.
Skin Protection	
Hand Protection:	Chemical resistant gloves.
Skin and Body Protection:	Wear appropriate clothing to prevent any possibility of skin contact. Wear work clothes with long sleeves and pants. If splashing or contact with liquid material is possible, consider the need for an impervious overcoat. Fire resistant (i.e., Nomex) or natural fiber clothing (i.e., cotton or wool) is recommended. Synthetic clothing can generate static electricity and is not recommended where a flammable vapor release may occur. Wear chemical-resistant safety footwear with good traction to prevent slipping. Static Dissipative (SD) rated footwear is also recommended.
Respiratory Protection:	Appropriate NIOSH approved air-purifying respirator or self-contained breathing apparatus should be used. Supplied air breathing apparatus must be used when oxygen concentrations are low or if airborne concentrations exceed the limits of the air-purifying respirators or IDLH levels.
Hygiene measures:	Use effective control measures and PPE to maintain worker exposure to concentrations that are below these limits. Ensure that eyewash stations and safety showers are in close proximity to work locations.

9. Physical and chemical properties

Appearance

Physical state:	liquid
Form:	liquid
Color:	Clear
Odor:	slight hydrocarbon
Odor Threshold:	No data available.
Melting point/freezing point:	No data available.
Initial boiling point and boiling range:	30 - 40 °C (86 - 104 °F)
Flammability (solid, gas):	Not applicable

Upper/lower limit on flammability or explosive limits

Flammability Limit - Upper (%):	10 %(V) (isoprene)
Flammability Limit - Lower (%):	1.4 %(V) (pentane)
Flash Point:	-51 °C (-60 °F)
Auto-ignition temperature:	220 °C (428 °F)
Decomposition temperature:	No data available.
pH:	Not applicable
Kinematic viscosity:	No data available.
Solubility(ies)	
Solubility in water:	690 mg/l (1,3-Pentadiene) (estimated)
Solubility (other):	No data available.
Partition coefficient (n-octanol/water):	1.5 (1,3-Pentadiene) (estimated)
Vapor pressure:	108 kPa (37.8 °C (100.0 °F)) (Reid Vapor Pressure)
Evaporation rate:	28.6 (pentane) n-butyl acetate=1
Density:	675 - 705 kg/m ³
Relative density:	0.675 - 0.705 (15 °C (59 °F)) (Water=1)
Vapor density:	2.5 (0 °C (32 °F)) (Air=1)
Particle characteristics	
Particle Size:	Not applicable
Other information	
Explosive properties:	No data available.

10. Stability and reactivity

Reactivity:	May attack and degrade some types of plastics, rubbers and coatings. Residues (hydrocarbons in polymer buildup) will react with air and may be a serious fire hazard. In the presence of air, explosive peroxides may be produced. Product may react (exothermic, ignite, polymerize) with some absorbents. Test with small quantity prior to using larger amounts. Strong oxidizers can increase fire and explosion hazard.
Chemical Stability:	Material is stable when inhibited and handled and stored properly.
Possibility of hazardous reactions:	Hazardous reactions can occur if product is in contact with oxygen.
Conditions to avoid:	Exposure to open flame or excessive heat can cause fire or explosion. Keep away from heat, sparks and open flame.
Incompatible Materials:	Oxidizing agents. Reducing agents. Acids.
Hazardous Decomposition Products:	Upon decomposition, this product emits carbon monoxide, carbon dioxide, low molecular weight hydrocarbons.

11. Toxicological information**Information on likely routes of exposure**

Inhalation:	Excessive inhalation of this product may result in central nervous system effects including headache, dizziness and nausea. Minute amounts aspirated into the lungs during ingestion or vomiting may cause severe pulmonary injury.
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Ingestion:	Harmful if swallowed. Minute amounts aspirated into the lungs during ingestion or vomiting may cause severe pulmonary injury. Ingestion of this product may result in vomiting, abdominal pain and central nervous system effects including headache, dizziness and nausea.
Skin Contact:	Causes skin irritation.
Eye contact:	Causes serious eye irritation.

Symptoms related to the physical, chemical and toxicological characteristics

Inhalation:	Central nervous system effects including headache, dizziness and nausea.
Ingestion:	Vomiting, abdominal pain and central nervous system effects including headache, dizziness and nausea.
Skin Contact:	Skin irritation.
Eye contact:	Eye irritation.

Information on toxicological effects**Acute toxicity (list all possible routes of exposure)**

Oral	
Product:	ATEmix: 1,292.07 mg/kg Harmful if swallowed.
Dermal	
Product:	Not classified for acute toxicity based on available data.
Inhalation	
Product:	Not classified for acute toxicity based on available data.

Repeated dose toxicity	
Product:	No data available.

Skin Corrosion/Irritation	
Product:	Causes skin irritation.

Serious Eye Damage/Eye Irritation	
Product:	Causes serious eye irritation.

Respiratory or Skin Sensitization	
Product:	No data available.

Components:	
4,7-Methano-1H-indene, 3a,4,7,7a-tetrahydro-	Skin sensitization, Draize (Guinea Pig): Not a skin sensitizer.

Carcinogenicity	
Product:	May cause cancer.

IARC Monographs on the Evaluation of Carcinogenic Risks to Humans:	
1,3-Butadiene, 2-methyl-	Overall evaluation: 2B. Possibly carcinogenic to humans.

ACGIH Carcinogen List:	
	No carcinogenic components identified

US. National Toxicology Program (NTP) Report on Carcinogens:	
1,3-Butadiene, 2-methyl-	Reasonably Anticipated to be a Human Carcinogen.

US. OSHA Specifically Regulated Substances (29 CFR 1910.1001-1053), as amended:

No carcinogenic components identified

Germ Cell Mutagenicity**In vitro****Product:** Suspected of causing genetic defects.**In vivo****Product:** Suspected of causing genetic defects.**Reproductive toxicity****Product:** No data available.**Components:**

4,7-Methano-1H-indene, Not classified.

3a,4,7,7a-tetrahydro-

Specific Target Organ Toxicity - Single Exposure**Product:** May cause drowsiness or dizziness.**Specific Target Organ Toxicity - Repeated Exposure****Product:** No data available.**Components:**

1,3-Pentadiene Nervous System

1,3-Butadiene, 2-methyl- Nervous System, Lungs

1,3-Cyclopentadiene Liver, Kidney

n-Pentane Nervous System - Chronic pentane exposure may damage the nervous system causing numbness, "pins and needles" and weakness in the arms and legs.

Skin - Prolonged and repeated skin contact can cause defatting dermatitis with dryness, cracking, redness and blisters.

Cyclopentane Nervous System - Chronic pentane exposure may damage the nervous system causing numbness, "pins and needles" and weakness in the arms and legs.

Skin - Prolonged and repeated skin contact can cause defatting dermatitis with dryness, cracking, redness and blisters.

Butane, 2-methyl- Nervous System - Chronic pentane exposure may damage the nervous system causing numbness, "pins and needles" and weakness in the arms and legs.

Skin - Prolonged and repeated skin contact can cause defatting dermatitis with dryness, cracking, redness and blisters.

Aspiration Hazard**Product:** May be fatal if swallowed and enters airways.**Other effects:** No data available.

12. Ecological information

Ecotoxicity:

Acute hazards to the aquatic environment:

Fish	
Product:	Harmful to aquatic life.
Aquatic Invertebrates	
Product:	Harmful to aquatic life.
Toxicity to Aquatic Plants	
Product:	Harmful to aquatic life.

Chronic hazards to the aquatic environment:

Fish	
Product:	Harmful to aquatic life with long lasting effects.
Aquatic Invertebrates	
Product:	Harmful to aquatic life with long lasting effects.
Toxicity to Aquatic Plants	
Product:	Harmful to aquatic life with long lasting effects.

Persistence and Degradability

Biodegradation	
Product:	No data available.
BOD/COD Ratio	
Product:	No data available.

Bioaccumulative potential

Bioconcentration Factor (BCF)	
Product:	No data available.
Components:	
4,7-Methano-1H-indene, 3a,4,7,7a-tetrahydro-	Carp, Bioconcentration Factor (BCF): 58.9 - 384
Partition Coefficient n-octanol / water (log Kow)	
Product:	1.5 (1,3-Pentadiene) (estimated)

Mobility in soil: Limited absorption into soil and sediment.

Other adverse effects: No data available.

13. Disposal considerations

Disposal instructions: Dispose of contents and container in accordance with local regulations. Waste generator is advised to carefully consider hazardous properties and control measures needed for other materials that may be found in the waste.

Contaminated Packaging: Check local, federal and state environmental regulations prior to disposal.

14. Transport information**DOT**

UN number or ID number:	UN 3295
UN Proper Shipping Name:	Hydrocarbons, liquid, n.o.s.
Transport Hazard Class(es)	
Class:	3
Label(s):	3
Packing Group:	I
Marine Pollutant:	No
Special precautions for user:	Reference Emergency Response Guidebook No. 128, latest revision.
Reportable quantity	1,3-Pentadiene 100 lbs 1-Pentene 100 lbs

15. Regulatory information**US Federal Regulations****TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D)**

<u>Chemical Identity</u>	<u>Reportable quantity</u>
1,3-Cyclopentadiene	De minimis concentration: 1.0% Subject to One-Time Reporting Requirements (Per Country)

US. Toxic Substances Control Act (TSCA) Section 5(a)(2) Final Significant New Use Rules (SNURs) (40 CFR 721 and 725, Subpt E)

None present or none present in regulated quantities.

US. OSHA Specifically Regulated Substances (29 CFR 1910.1001-1053), as amended

None present or none present in regulated quantities.

CERCLA Hazardous Substance List (40 CFR 302.4):

<u>Chemical Identity</u>	<u>Name on List:</u>	<u>Reportable quantity</u>
1,3-Pentadiene	1-METHYLBUTADIENE	100 lbs.
Cyclopentene	RCRA Hazardous Waste No. D001	100 lbs.
1-Pentene	RCRA Hazardous Waste No. D001	100 lbs.
1,3-Butadiene, 2-methyl-	ISOPRENE	100 lbs.
1-Butene, 2-methyl-	RCRA Hazardous Waste No. D001	100 lbs.
n-Pentane	RCRA Hazardous Waste No. D001	100 lbs.
Cyclopentane	RCRA Hazardous Waste No. D001	100 lbs.
2-Butene, 2-methyl-	RCRA Hazardous Waste No. D001	100 lbs.
Butane, 2-methyl-	RCRA Hazardous Waste No. D001	100 lbs.

Superfund Amendments and Reauthorization Act of 1986 (SARA)**Hazard categories**

Flammable (gases, aerosols, liquids, or solids), Acute toxicity (any route of exposure), Skin Corrosion or Irritation, Serious eye damage or eye irritation, Germ Cell Mutagenicity, Carcinogenicity, Specific target organ toxicity (single or repeated exposure), Aspiration Hazard, Hazards Not Otherwise Classified (HNOC)

US. EPCRA (SARA Title III) Section 304 Extremely Hazardous Substances Reporting Quantities and the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) Hazardous Substances

None present or none present in regulated quantities.

US. EPCRA (SARA Title III) Section 313 Toxic Chemical Release Inventory (TRI) Reporting

<u>Chemical Identity</u>	<u>Reporting threshold for other users</u>	<u>Reporting threshold for manufacturing and processing</u>
1,3-Butadiene, 2-methyl-	10000 lbs	25000 lbs.

4,7-Methano-1H-indene, 3a,4,7,7a-tetrahydro-	10000 lbs	25000 lbs.
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Clean Air Act (CAA) Section 112(r) Accidental Release Prevention (40 CFR 68.130):

<u>Chemical Identity</u>	<u>Reportable quantity</u>
1,3-Pentadiene	10000 lbs
1-Pentene	10000 lbs
1,3-Butadiene, 2-methyl-	10000 lbs
1-Butene, 2-methyl-	10000 lbs
n-Pentane	10000 lbs
Butane, 2-methyl-	10000 lbs

Clean Water Act Section 311 Hazardous Substances (40 CFR 117.3)

<u>Chemical Identity</u>	<u>Reportable quantity</u>
1,3-Butadiene, 2-methyl-	100 lbs.
Benzene	10 lbs.

US State Regulations**US. California Proposition 65**

WARNING: This product can expose you to chemicals including, Benzene; which is [are] known to the State of California to cause cancer and birth defects or other reproductive harm.

This product can expose you to chemicals including, 1,3-Butadiene, 2-methyl-; which is [are] known to the State of California to cause cancer.

For more information go to www.P65Warnings.ca.gov.

Inventory Status

Canada DSL Inventory List:	On or in compliance with the inventory
US TSCA Inventory:	On or in compliance with the inventory

16. Other information, including date of preparation or last revision

Issue Date: 02/13/2026

Revision Information: 02/13/2026: SDS Update – phrase edits
07/23/2024: SDS Update – OEL updates and phrase edits
02/23/2024: SDS Update – GHS classification change, composition edits, OEL edits, section 9 edits, section 11 edits, section 15 edits, phrase edits
03/02/2022: SDS Update – address updated
01/02/2020: SDS Update

Version #: 9.2

Abbreviations and acronyms:

ACC = American Chemistry Council; ACGIH = American Conference of Governmental Industrial Hygienists; BOD = Biochemical Oxygen Demand; C = Ceiling; CAS = Chemical Abstracts Service; CERCLA = Comprehensive Environmental Response, Compensation, and Liability Act; CFR = Code of Federal Regulations; COD = Chemical Oxygen Demand; DOT = Department of Transportation; DSL = Domestic Substances List; EC50 = Effective Concentration 50%; EPA = Environmental Protection Agency; GHS = Globally Harmonized System for the Classification and Labelling of Chemicals; HPV = High Production Volume; IARC = International Agency for Research on Cancer; LC50 = Lethal Concentration 50%; LD50 = Lethal Dose 50%; NFPA = National Fire Protection Association; NIOSH = National Institute for Occupational Safety and Health; NTP = National Toxicology Program; OSHA = Occupational Safety and Health Administration; PEL = Permissible Exposure Limit; PMCC = Pensky-Martens Closed Cup; PPE = Personal Protective Equipment; RCRA = Resource Conservation and Recovery Act; REL = Recommended Exposure Limit; SARA = Superfund Amendments and Reauthorization Act; SCBA = Self Contained Breathing Apparatus; SDS = Safety Data Sheet; STEL = Short Term Exposure Limit; TLV = Threshold Limit Value; TSCA = Toxic Substances Control Act; TWA = Time Weighted Average

Further Information:

For additional information on equipment bonding and grounding, refer to the American Petroleum Institute (API) Recommended Practice 2003, "Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents" or National Fire Protection Association (NFPA) 77, "Recommended Practice on Static Electricity".

For additional information on storing and handling flammable liquids, refer to the National Fire Protection Association (NFPA) 30, "Flammable and Combustible Liquids Code".

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